

AI: System Architecture

Core pattern — every module: Sonar (1-2 line question → live web retrieval) → GPT (structure / extract / score)

Sonar is for retrieval only. GPT never fetches — it processes what Sonar returned. BaseModule provides retry, deflection detection, and structured extraction to all modules.

Part 1 — Input Pipeline

Entry point: `run_pipeline(url)` → `PipelineResult` · 5 sequential stages

Stage	What it does
Crawl	Playwright BFS crawler, depth 3 / 500 pages. URLs classified as product / category / about / contact / blog / ignore with type-specific depth budgets.
Product Extraction	GPT-4o-mini (concurrency=3). Single-product and tile-grid modes. Deduplicated by <code>product_name</code> .
Company Extraction	Single GPT call across all about/contact pages → <code>CompanyData</code> (name, location, certs, business type).
Confidence Scoring	Rule-based 0–100 score. High ≥85 (ready), Medium 40–84 (review), Low <40 (noise).
Semantic Scoring	Optional. GPT re-evaluates Medium-band products only — skips clear High/Low to save tokens.

Part 2 — Intelligence Engine

Entry point: `ModuleRunner().run_all(inp)` → `RunnerResult` · 8 modules, parallel execution

Module	Output	Key Design Note
Market Demand	7 fields per country across 15 markets	Parallel per country; semaphore(3)
Buyer Discovery	Top 10 buyer companies in target country	3 calls: Sonar → extract → dedup
Competitor Discovery	Top 10 global manufacturers/exporters	3 calls; extensible for intel module
Variants & Formats	6 variants: price, MOQ, demand, lead time	2 Sonar calls: discover → enrich
Price Analysis	Market range, margin %, cert premiums	Depends on Variants output
Keyword Intelligence	Buyer-intent, gap, multilingual keyword buckets	Google Ads API + GPT classification
Email Sequence	3-email cold outreach sequence	GPT-only, no Sonar
Ad Concepts	6 creative B2B ad concepts	GPT-only, temp=0.9, dynamic angles

Key Design Decisions

• Never-raise contract: every run() returns typed result with success=False — runner always gets a usable object. • Sonar deflection retry: 2+ deflection phrases trigger a single retry. Same query usually succeeds on retry. • Variants 2-call split: discover names first, then fetch market data for that known list — richer results.	• Keywords batch API: all seeds in one Google Ads call, not per-seed — avoids 429 rate limits. • PriceAnalysis dependency: only module that takes another module's output (Variants) — runs after it in the runner. • Short Sonar prompts: 1-2 line direct questions outperform long structured prompts — retrieval degrades with context.
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